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GROUP SPECIFICATION

Fifth Generation Fixed Network (F5G); Test Specification for fgOTN Functionality and Performance

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ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
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Foreword

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Modal verbs terminology

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1 Scope

The present document describes the test methodology and criteria for the functionality and performance of ITU-T G.709-based fine grain Optical Transport Network (fgOTN). The test specification includes fine grain flexible Optical Data Unit (fgODUflex) path layer, multiplexing, hitless bandwidth adjustment, protection and clock synchronization in transport scenarios.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] Recommendation ITU-T G.709: "Interfaces for the optical transport network".

[2] Recommendation ITU-T G.709.20: "Overview of fine grain OTN".

[3] Recommendation ITU-T G.798.3: "Characteristics of optical transport network hierarchy equipment functional blocks - OTU and ODU".

[4] Recommendation ITU-T G.808.1: "Generic protection switching - Linear trail and subnetwork protection".

[5] Recommendation ITU-T G.808.4: "Linear protection for fine grain metro transport network (fgMTN) and fine grain optical transport network (fgOTN)".

[6] Recommendation ITU-T G.825: "The control of jitter and wander within digital networks which are based on the synchronous digital hierarchy (SDH)".

[7] Recommendation ITU-T G.8251: "The control of jitter and wander within the optical transport network (OTN)".

2.2 Informative references

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[i.1] Recommendation ITU-T G.808: "Terms and definitions for network protection and restoration".

[i.2] Recommendation ITU-T G.810: "Definitions and terminology for synchronization networks".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

1+1: protection architecture with one normal traffic signal, one working transport entity, one protection transport entity and a permanent bridge

NOTE 1: At the source end, the normal traffic signal is permanently bridged to both the working and the protection transport entities. At the sink end, the normal traffic signal is selected from the better of the two transport entities.

NOTE 2: Due to the permanent bridging, the 1+1 protection architecture does not allow an unprotected extra traffic signal to be provided.

NOTE 3: This definition is derived from the term "1+1 (protection) architecture" in Recommendation ITU-T G.808 [i.1].

jitter: short-term variations of the significant instants of a timing signal from their ideal positions in time

NOTE 1: Short-term implies that these variations are of frequency greater than or equal to 10 Hz.

NOTE 2: This definition is derived from Recommendation ITU-T G.810 [i.2].

wander: long-term variations of the significant instants of a digital signal from their ideal position in time

NOTE 1: Long-term implies that these variations are of frequency less than 10 Hz.

NOTE 2: This definition is derived from Recommendation ITU-T G.810 [i.2].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

AIS	Alarm Insertion Signal
BDI	Backward Defect Indication
BEI	Backward Error Indication
BIP	Bit Interleaved Parity
CBR	Constant Bit Rate
CSF	Client Signal Fail
DM	Delay Measurement
DUT	Device Under Test
E1	2 048 kbit/s PDH signal

NOTE: The term E1 refers to an E-carrier level 1. In ITU-T the electrical 2 048 kbit/s PDH signal is referred to as an E12 while the 2 048 kbit/s PDH end-to-end service provided by the fgOTN network is referred to as a P12x transparent PDH path of level 12. In the remainder of the present document the term E1 is used.

EBC	Errored Block Count
EMS	Equipment Management System
F_EBC	Far-end EBC
FAS	Frame Alignment Signal
fgODUflex	fine grain flexible Optical Data Unit
fgOTN	fine grain Optical Transport Network
IAE	Incoming Alignment Error

LCK	Locked
LOF	Loss Of Frame
LOFLOM	Loss Of Frame and Loss Of Multiframe
LOM	Loss Of Multiframe
LOS	Loss Of Signal
LTC	Loss of Tandem Connection
MFAS	MultiFrame Alignment Signal
N_EBC	Near-end EBC
NE	Network Element
NR	No Request
OCI	Open Connection Indication
ODCp	ODU Clock, payload
ODUj	Optical Data Unit of level j
ODUk	Optical Data Unit of level k
OTUk	Optical Transport Unit of level k
PDH	Plesiochronous Digital Hierarchy
PHY	Physical layer device
PLM	Payload Mismatch
PM	Path Monitoring
PT	Payload Type
SD	Signal Degrade
SDH	Synchronous Digital Hierarchy
SF	Signal Fail
SNC	SubNetwork Connection
SNC/I	SubNetwork Connection with Inherent monitoring
SNC/N	SubNetwork Connection with Non-intrusive monitoring
SNC/S	SubNetwork Connection with Sub-layer monitoring
SSF	Server Signal Fail
STM-N	Synchronous Transport Module, level N
TCM	Tandem Connection Monitoring
TDM	Time-Division Multiplexing
TIM	Trail trace Identifier Mismatch
ToD	Time of Day
TTI	Trail Trace Identifier
VC-n	Virtual Container, level n

4 Overview

For fgOTN, the test cases focus on the requirements from the Recommendations ITU-T. Table 1 shows fgOTN test cases including fgODUflex path layer function and performance, fgODUflex multiplexing function and performance, fgODUflex hitless bandwidth adjustment function and performance, fgODUflex protection function and performance, fgODUflex clock synchronization function and performance. fgOTN analysers used in test configurations in clause 5 below shall comply with the processes specified in Recommendation ITU-T G.798.3 [3].

Table 1: Test cases for fgOTN [2]

Clause	Test case	Recommendation ITU-T
5.1	Path layer function and performance	
5.1.1	fgOTN Test case #1.1: fgODUflex frame alignment	ITU-T G.709 [1], Annex M.2.1
5.1.2	fgOTN Test case #1.2: fgODUflex multiframe alignment	ITU-T G.709 [1], Annex M.2.1
5.1.3	fgOTN Test case #1.3: fgOPUflex Payload Type (PT)	ITU-T G.709 [1], Annex M.2.4
5.1.4	fgOTN Test case #1.4: fgODUflex Path Monitoring (PM) Trail Trace Identifier (TTI)	ITU-T G.709 [1], Annex M.2.2
5.1.5	fgOTN Test case #1.5: fgODUflex PM signal quality	ITU-T G.709 [1], Annex M.2.2
5.1.6	fgOTN Test case #1.6: fgODUflex PM maintenance signals	ITU-T G.709 [1], Annex M.3
5.1.7	fgOTN Test case #1.7: fgODUflex PM delay measurement	ITU-T G.709 [1], Annex M.2.2.6
5.1.8	fgOTN Test case #1.8: fgODUflex Tandem Connection Monitoring (TCM) Trail Trace Identifier (TTI)	ITU-T G.709 [1], Annex M.2.2
5.1.9	fgOTN Test case #1.9: fgODUflex TCM signal quality	ITU-T G.709 [1], Annex M.2.2
5.1.10	fgOTN Test case #1.10: fgODUflex TCM maintenance signals	ITU-T G.709 [1], Annex M.3
5.1.11	fgOTN Test case #1.11: fgODUflex TCM delay measurement	ITU-T G.709 [1], Annex M.2.2.6
5.2	fgOTN client services and performance	
5.2.1	fgOTN Test case #2.1: STM-N clients	ITU-T G.709 [1], Annex M.5.3
5.2.2	fgOTN Test case #2.2: STM-N delay and jitter/wander	ITU-T G.8251 [7]
5.2.3	fgOTN Test case #2.3: Ethernet clients	ITU-T G.709 [1], Annex M.5.2
5.2.4	fgOTN Test case #2.4: Ethernet delay	ITU-T G.709 [1], Annex M.2.2.6
5.2.5	fgOTN Test case #2.5: VC-n clients	ITU-T G.709 [1], Annex M.5.4.1
5.2.6	fgOTN Test case #2.6: E1 clients	ITU-T G.709 [1], Annex M.5.4.2
5.3	Protection function and performance	
5.3.1	fgOTN Test case #3.1: 1+1 SNC/I protection	ITU-T G.808.1 [4]
5.3.2	fgOTN Test case #3.2: 1+1 SNC/N protection	ITU-T G.808.1 [4]
5.3.3	fgOTN Test case #3.3: 1+1 SNC/S protection	ITU-T G.808.4 [5]
5.4	Hitless bandwidth adjustment	
5.4.1	fgOTN Test case #4.1: Hitless bandwidth adjustment without protection	ITU-T G.709 [1], Annex O.1
5.4.2	fgOTN Test case #4.2: Hitless bandwidth adjustment in protection scenarios	ITU-T G.709 [1], Annex O.1
5.5	Cross-connect	
5.5.1	fgOTN Test case #5.1: Cross-connect capacity	
5.5.2	fgOTN Test case #5.2: Cross-connect delay	ITU-T G.709 [1], Annex M.2.2.6
5.5.3	fgOTN Test case #5.3: fgODUflex container capacity	ITU-T G.709 [1], Annex N.1

5 Test cases

5.1 fgOTN path layer function and performance test cases

5.1.1 fgOTN Test case #1.1: fgODUflex frame alignment

5.1.1.1 Test purpose

This test case verifies fgODUflex frame alignment.

5.1.1.2 Test configuration

The test configuration is shown in Figure 1.

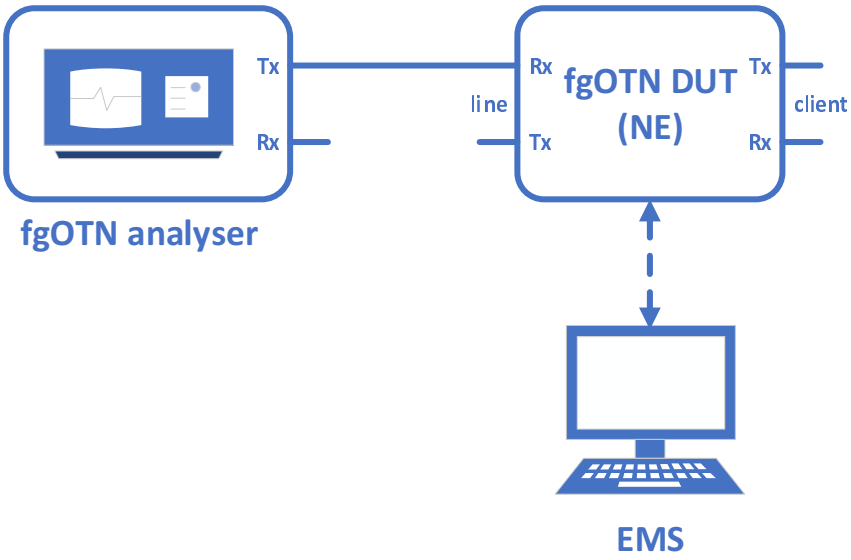


Figure 1: fgODUflex alignment test configuration

The test configuration in Figure 1 is reused in multiple test cases below.

5.1.1.3 Test procedure

The test procedure is as follows:

1) Connect the Device Under Test (DUT) according to the test configuration in Figure 1.
2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT.
3) Continuously insert an errored Frame Alignment Signal (FAS) on the fgODUflex path on the fgOTN analyser. The correct FAS pattern is specified in Recommendation ITU-T G.709 [1].
4) Stop inserting the errored FAS on the fgOTN analyser.

5.1.1.4 Expected result

In step 2, the fgOTN analyser shall not show any errors. The EMS shall not show any errors.

In step 3, the EMS shall raise a Loss Of Frame (LOF) alarm on the configured fgODUflex.

In step 4, the EMS shall clear the LOF alarm on the configured fgODUflex.

5.1.2 fgOTN Test case #1.2: fgODUflex multiframe alignment

5.1.2.1 Test purpose

This test case verifies fgODUflex multiframe alignment.

5.1.2.2 Test configuration

The test configuration is shown in Figure 1 above.

5.1.2.3 Test procedure

The test procedure is as follows:

1) Connect the Device Under Test (DUT) according to the test configuration in Figure 1.
2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT.
3) Continuously insert an errored Multiframe Alignment Signal (MFAS) on the fgODUflex path on the fgOTN analyser. The correct MFAS pattern is specified in Recommendation ITU-T G.709 [1].
4) Stop inserting the errored MFAS on the fgOTN analyser.

5.1.2.4 Expected result

In step 2, the fgOTN analyser shall not show any errors. The EMS shall not show any errors.

In step 3, the EMS shall raise a Loss Of Multiframe (LOM) alarm on the configured fgODUflex.

In step 4, the EMS shall clear the LOM alarm on the configured fgODUflex.

5.1.3 fgOTN Test case #1.3: fgOPUflex Payload Type (PT)

5.1.3.1 Test purpose

This test case verifies fgODUflex Payload Type (PT).

5.1.3.2 Test configuration

The test configuration is shown in Figure 2.

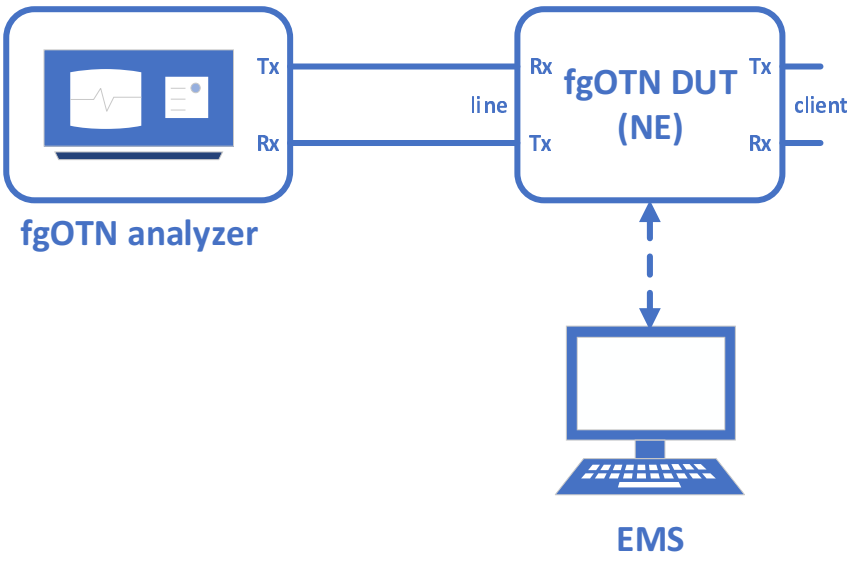


Figure 2: fgODUflex overhead test configuration

The test configuration in Figure 2 is reused in multiple test cases below.

5.1.3.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 2.
2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT.

3) Configure CBR over fgODUflex service on both the DUT and the fgOTN analyser.

4) On the fgOTN analyser: modify the inserted fgODUflex PT value or configure a different client over fgODUflex service.

5) On the fgOTN analyser: restore the inserted fgODUflex PT value or restore the CBR over fgODUflex configuration.

6) Configure a different client over fgODUflex service on the DUT. Read the received PT value on the fgOTN analyser.

7) Restore the CBR over fgODUflex configuration on the DUT. Read the received PT value on the fgOTN analyser.

5.1.3.4 Expected result

In step 3, the fgOTN analyser shall not show any errors. The EMS shall not show any errors.

In step 4, the EMS shall raise a Payload Mismatch (PLM) alarm on the configured fgODUflex.

In step 5, the EMS shall clear the PLM alarm on the configured fgODUflex.

In step 6, the fgOTN analyser shall display the PT value corresponding to the chosen client over fgODUflex service.

In step 7, the fgOTN analyser shall display the PT value corresponding to the CBR over fgODUflex service.

NOTE: fgODUflex PT values are specified in Recommendation ITU-T G.709 [1].

5.1.4 fgOTN Test case #1.4: fgODUflex Path Monitoring (PM) Trail Trace Identifier (TTI)

5.1.4.1 Test purpose

This test case verifies fgODUflex end-to-end connectivity using the PM TTI.

5.1.4.2 Test configuration

The test configuration is shown in Figure 2 above.

5.1.4.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 2.

2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT.

3) Configure the transmitted PM TTI on the fgOTN analyser and the expected PM TTI on the DUT to the same TTI string.

4) On the fgOTN analyser: modify the transmitted PM TTI string.

5) On the fgOTN analyser: restore the transmitted PM TTI string to the original TTI string in step 3.

6) Configure the transmitted PM TTI on the DUT. Read the received PM TTI string on the fgOTN analyser.

5.1.4.4 Expected result

In step 3, the fgOTN analyser shall not show any errors. The EMS shall not show any errors. The EMS shall report the TTI string inserted on the fgOTN analyser.

In step 4, the EMS shall raise a Trail trace Identifier Mismatch (TIM) alarm on the PM layer of the configured fgODUflex. The EMS shall report the modified TTI string inserted on the fgOTN analyser. The fgOTN analyser shall raise a Backward Defect Indication (BDI) alarm on the PM layer of the configured fgODUflex.

In step 5, the EMS shall clear the TIM alarm on PM layer of the configured fgODUflex. The EMS shall report the original TTI string inserted on the fgOTN analyser. The fgOTN analyser shall clear the BDI alarm on the PM layer of the configured fgODUflex.

In step 6, the fgOTN analyser shall display the TTI string inserted on the DUT.

NOTE: The format of the TTI string is specified in Recommendation ITU-T G.709 [1]. For the purpose of this test case the TTI string can be random.

5.1.5 fgOTN Test case #1.5: fgODUflex PM signal quality

5.1.5.1 Test purpose

This test case verifies fgODUflex PM signal quality.

5.1.5.2 Test configuration

The test configuration is shown in Figure 2 above.

5.1.5.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 2.
2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT.
3) On the fgOTN analyser: insert Bit Interleaved Parity (BIP-8) errors on the PM layer.
4) On the fgOTN analyser: insert Backward Error Indication (BEI) errors on the PM layer.

5.1.5.4 Expected result

In step 2, the fgOTN analyser shall not show any errors. The EMS shall not show any errors.

In step 3, the EMS shall report PM Near-end Errored Block Counts (N_EBC) corresponding to the inserted BIP-8 errors. The fgOTN analyser shall display BEI counts equal to the inserted BIP-8 error count.

In step 4, the EMS shall report PM Far-end Errored Block Counts (F_EBC) corresponding to the inserted BEI errors.

5.1.6 fgOTN Test case #1.6: fgODUflex PM maintenance signals

5.1.6.1 Test purpose

This test case verifies fgODUflex PM maintenance signals and fgODUflex end-to-end continuity.

5.1.6.2 Test configuration

The test configuration is shown in Figure 2 above.

5.1.6.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 2.
2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT.

3) On the fgOTN analyser: insert the administrative Lock (LCK) maintenance signal on the PM layer.

4) On the fgOTN analyser: clear the LCK insertion on the PM layer.

5) On the fgOTN analyser: insert Open Connection Indication (OCI) on the PM layer.

6) On the fgOTN analyser: clear the OCI insertion on the PM layer.

7) On the fgOTN analyser: insert Alarm Indication Signal (AIS) on the PM layer.

8) On the fgOTN analyser: clear the AIS insertion on the PM layer.

9) On the fgOTN analyser: insert BDI on the PM layer.

10) On the fgOTN analyser: clear the BDI insertion on the PM layer.

11) On the EMS: administratively lock the PM of the configured fgODUflex.

12) On the EMS: clear the administrative lock.

In case the DUT supports fgODUflex cross-connection steps 13 to 17 shall be performed:

13) On the EMS: configure a fgODUflex connection from the receive OTN line port connected to the fgOTN analyser to the transmit OTN line port connected to the fgOTN analyser.

14) On the EMS: remove the connection between receive and transmit OTN line port.

15) On the EMS: restore the connection between receive and transmit OTN line port.

16) Disconnect the fibre between fgOTN analyser and the OTN line port on the DUT.

17) Reconnect the fibre between fgOTN analyser and the OTN line port on the DUT.

5.1.6.4 Expected result

In step 2, the fgOTN analyser shall not show any errors. The EMS shall not show any errors.

In step 3, the EMS shall report LCK alarm on the PM layer of the configured fgODUflex. The fgOTN analyser shall raise a BDI alarm on the PM layer of the configured fgODUflex.

In step 4, the EMS shall clear the LCK alarm on PM layer of the configured fgODUflex. The fgOTN analyser shall clear the BDI alarm on the PM layer of the configured fgODUflex.

In step 5, the EMS shall report OCI alarm on the PM layer of the configured fgODUflex. The fgOTN analyser shall raise a BDI alarm on the PM layer of the configured fgODUflex.

In step 6, the EMS shall clear the OCI alarm on PM layer of the configured fgODUflex. The fgOTN analyser shall clear the BDI alarm on the PM layer of the configured fgODUflex.

In step 7, the EMS shall report AIS alarm on the PM layer of the configured fgODUflex. The fgOTN analyser shall raise a BDI alarm on the PM layer of the configured fgODUflex.

In step 8, the EMS shall clear the AIS alarm on PM layer of the configured fgODUflex. The fgOTN analyser shall clear the BDI alarm on the PM layer of the configured fgODUflex.

In step 9, the EMS shall report BDI alarm on the PM layer of the configured fgODUflex.

In step 10, the EMS shall clear the BDI alarm on PM layer of the configured fgODUflex.

In step 11, the fgOTN analyser shall report LCK alarm on the PM layer of the configured fgODUflex.

In step 12, the fgOTN analyser shall clear the LCK alarm on PM layer of the configured fgODUflex.

In case the DUT supports fgODUflex cross-connection the following results are expected:

In step 13, the fgOTN analyser shall not show any errors.

In step 14, the DUT shall insert the OCI maintenance signal which the fgOTN analyser shall detect and report as an OCI alarm on the PM layer of the configured fgODUflex.

In step 15, the fgOTN analyser shall clear the OCI alarm on PM layer of the configured fgODUflex.

In step 16, the EMS shall report LOS alarm on the OTN line port. The DUT shall insert the AIS maintenance signal which the fgOTN analyser shall detect and report as an AIS alarm on the PM layer of the configured fgODUflex.

In step 17, the EMS shall clear the LOS alarm on the OTN line port. The fgOTN analyser shall clear the AIS alarm on PM layer of the configured fgODUflex.

5.1.7 fgOTN Test case #1.7: fgODUflex PM Delay Measurement (DM)

5.1.7.1 Test purpose

This test case verifies bi-directional fgODUflex PM delay measurement.

NOTE 1: Bi-directional DM does not require Time of Day (ToD) synchronization between fgOTN analyser and DUT.

Uni-directional fgODUflex PM delay measurement is for further study.

NOTE 2: Uni-directional DM requires ToD synchronization between fgOTN analyser and DUT.

5.1.7.2 Test configuration

The test configuration is shown in Figure 3.

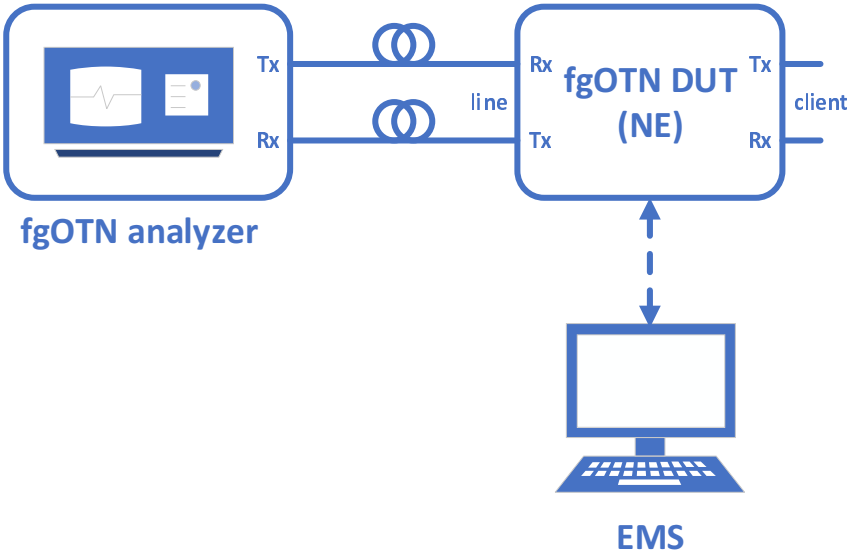


Figure 3: fgODUflex delay measurement test configuration

The test configuration in Figure 3 is reused in multiple test cases below. The difference to Figure 2 is the fibre length between the fgOTN analyser and the DUT, which shall be longer than 1 km.

5.1.7.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 3. The fibre between fgOTN analyser and DUT is longer than 1 km and identical in both directions.

2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT.

3) On the EMS: request a bi-directional DM on PM layer of the configured fgODUflex.

4) On the fgOTN analyser: request a bi-directional DM on PM layer of the configured fgODUflex.

5.1.7.4 Expected result

In step 3, the EMS shall report a delay value T_1.

In step 4, the fgOTN analyser shall report a delay value T_2. Verify that T_1 equals T_2.

5.1.8 fgOTN Test case #1.8: fgODUflex Tandem Connection Monitoring (TCM) Trail Trace Identifier (TTI)

5.1.8.1 Test purpose

This test case verifies fgODUflex subnetwork connectivity using the TCM TTI.

5.1.8.2 Test configuration

The test configuration is shown in Figure 2 above.

5.1.8.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 2.

2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT. Configure TCM monitoring at level TCMi to OPERATIONAL. Select the same TCMi level on fgOTN analyser and DUT.

3) Configure the transmitted TCMi TTI on the fgOTN analyser and the expected TCMi TTI on the DUT to the same TTI string.

4) On the fgOTN analyser: modify the transmitted TCMi TTI string.

5) On the fgOTN analyser: restore the transmitted TCMi TTI string to the original TTI string in step 3.

6) Configure the transmitted TCMi TTI on the DUT. Read the received TCMi TTI string on the fgOTN analyser.

NOTE: fgODUflex supports two TCM levels TCMi.

Repeat the test for each TCMi level supported by the DUT.

5.1.8.4 Expected result

In step 3, the fgOTN analyser shall not show any errors. The EMS shall not show any errors. The EMS shall report the TTI string inserted on the fgOTN analyser.

In step 4, the EMS shall raise a TIM alarm on the TCMi sublayer of the configured fgODUflex. The EMS shall report the modified TTI string inserted on the fgOTN analyser. The fgOTN analyser shall raise a Backward Defect Indication (BDI) alarm on the TCMi sublayer of the configured fgODUflex.

In step 5, the EMS shall clear the TIM alarm on TCMi sublayer of the configured fgODUflex. The EMS shall report the original TTI string inserted on the fgOTN analyser. The fgOTN analyser shall clear the BDI alarm on the TCMi sublayer of the configured fgODUflex.

In step 6, the fgOTN analyser shall display the TTI string inserted on the DUT.

NOTE: The format of the TTI string is specified in Recommendation ITU-T G.709 [1]. For the purpose of this test case the TTI string can be random.

5.1.9 fgOTN Test case #1.9: fgODUflex TCM signal quality

5.1.9.1 Test purpose

This test case verifies fgODUflex TCM signal quality.

5.1.9.2 Test configuration

The test configuration is shown in Figure 2 above.

5.1.9.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 2.

2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT. Configure TCM monitoring at level TCMi to OPERATIONAL. Select the same TCMi level on fgOTN analyser and DUT.

3) On the fgOTN analyser: insert BIP-8 errors on the TCMi sublayer.

4) On the fgOTN analyser: insert BEI errors on the TCMi sublayer.

NOTE: fgODUflex supports two TCM levels TCMi.

Repeat the test for each TCMi level supported by the DUT.

5.1.9.4 Expected result

In step 2, the fgOTN analyser shall not show any errors. The EMS shall not show any errors.

In step 3, the EMS shall report TCMi Near end Errored Block Counts (N_EBC) corresponding to the inserted BIP-8 errors. The fgOTN analyser shall display BEI counts equal to the inserted BIP-8 error count.

In step 4, the EMS shall report TCMi Far end Errored Block Counts (F_EBC) corresponding to the inserted BEI errors.

5.1.10 fgOTN Test case #1.10: fgODUflex TCM maintenance signals

5.1.10.1 Test purpose

This test case verifies fgODUflex TCM maintenance signals and fgODUflex subnetwork continuity.

NOTE: Forward and backward Incoming Alignment Error (IAE) maintenance signals are for further study.

5.1.10.2 Test configuration

The test configuration is shown in Figure 2 above.

5.1.10.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 2.

2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT. Configure TCM monitoring at level TCMi to OPERATIONAL. Select the same TCMi level on fgOTN analyser and DUT.

3) On the fgOTN analyser: insert the LCK maintenance signal on the TCMi sublayer.

4) On the fgOTN analyser: clear the LCK insertion on the TCMi sublayer.

5) On the fgOTN analyser: insert OCI on the TCMi sublayer.

6) On the fgOTN analyser: clear the OCI insertion on the TCMi sublayer.

7) On the fgOTN analyser: insert Loss of Tandem Connection (LTC) on the TCMi sublayer.

8) On the fgOTN analyser: clear the LTC insertion on the TCMi sublayer.

9) On the fgOTN analyser: insert AIS on the TCMi sublayer.

10) On the fgOTN analyser: clear the AIS insertion on the TCMi sublayer.

11) On the fgOTN analyser: insert BDI on the TCMi sublayer.

12) On the fgOTN analyser: clear the BDI insertion on the TCMi sublayer.

13) On the EMS: administratively lock the TCMi of the configured fgODUflex.

14) On the EMS: clear the administrative lock.

15) On the EMS: configure TCM monitoring at level TCMi to TRANSPARENT.

16) On the EMS: restore the TCM monitoring at level TCMi to OPERATIONAL.

In case the DUT supports fgODUflex cross-connection steps 17 to 21 shall be performed:

17) On the EMS: configure a fgODUflex connection from the receive OTN line port connected to the fgOTN analyser to the transmit OTN port connected to the fgOTN analyser.

18) On the EMS: remove the connection between receive and transmit OTN line port.

19) On the EMS: restore the connection between receive and transmit OTN line port.

20) Disconnect the fibre between fgOTN analyser and the OTN line port on the DUT.

21) Reconnect the fibre between fgOTN analyser and the OTN line port on the DUT.

NOTE: fgODUflex supports two TCM levels TCMi.

Repeat the test for each TCMi level supported by the DUT.

5.1.10.4 Expected result

In step 2, the fgOTN analyser shall not show any errors. The EMS shall not show any errors.

In step 3, the EMS shall report LCK alarm on the TCMi sublayer of the configured fgODUflex. The fgOTN analyser shall raise a BDI alarm on the TCMi sublayer of the configured fgODUflex.

In step 4, the EMS shall clear the LCK alarm on TCMi sublayer of the configured fgODUflex. The fgOTN analyser shall clear the BDI alarm on the TCMi sublayer of the configured fgODUflex.

In step 5, the EMS shall report OCI alarm on the TCMi sublayer of the configured fgODUflex. The fgOTN analyser shall raise a BDI alarm on the TCMi sublayer of the configured fgODUflex.

In step 6, the EMS shall clear the OCI alarm on TCMi sublayer of the configured fgODUflex. The fgOTN analyser shall clear the BDI alarm on the TCMi sublayer of the configured fgODUflex.

In step 7, the EMS shall report LTC alarm on the TCMi sublayer of the configured fgODUflex. The fgOTN analyser shall raise a BDI alarm on the TCMi sublayer of the configured fgODUflex.

In step 8, the EMS shall clear the LTC alarm on TCMi sublayer of the configured fgODUflex. The fgOTN analyser shall clear the BDI alarm on the TCMi sublayer of the configured fgODUflex.

In step 9, the EMS shall report AIS alarm on the TCMi sublayer of the configured fgODUflex. The fgOTN analyser shall raise a BDI alarm on the TCMi sublayer of the configured fgODUflex.

In step 10, the EMS shall clear the AIS alarm on TCMi sublayer of the configured fgODUflex. The fgOTN analyser shall clear the BDI alarm on the TCMi sublayer of the configured fgODUflex.

In step 11, the EMS shall report BDI alarm on the TCMi sublayer of the configured fgODUflex.

In step 12, the EMS shall clear the BDI alarm on TCMi sublayer of the configured fgODUflex.

In step 13, the fgOTN analyser shall report LCK alarm on the TCMi sublayer of the configured fgODUflex.

In step 14, the fgOTN analyser shall clear the LCK alarm on TCMi sublayer of the configured fgODUflex.

In step 15, the fgOTN analyser shall report LTC alarm on the TCMi sublayer of the configured fgODUflex.

In step 16, the fgOTN analyser shall clear the LTC alarm on TCMi sublayer of the configured fgODUflex.

In case the DUT supports fgODUflex cross-connection the following results are expected:

In step 17, the fgOTN analyser shall not show any errors.

In step 18, the DUT shall insert the OCI maintenance signal which the fgOTN analyser shall detect and report as an OCI alarm on the TCMi sublayer of the configured fgODUflex.

In step 19, the fgOTN analyser shall clear the OCI alarm on TCMi sublayer of the configured fgODUflex.

In step 20, the EMS shall report LOS alarm on the OTN line port. The DUT shall insert the AIS maintenance signal which the fgOTN analyser shall detect and report as an AIS alarm on the TCMi sublayer of the configured fgODUflex.

In step 21, the EMS shall clear the LOS alarm on the OTN line port. The fgOTN analyser shall clear the AIS alarm on TCMi sublayer of the configured fgODUflex.

5.1.11 fgOTN Test case #1.11: fgODUflex TCM Delay Measurement (DM)

5.1.11.1 Test purpose

This test case verifies bi-directional fgODUflex TCM delay measurement.

NOTE 1: Bi-directional DM does not require Time of Day (ToD) synchronization between fgOTN analyser and DUT.

Uni-directional fgODUflex TCM delay measurement is for further study.

NOTE 2: Uni-directional DM requires ToD synchronization between fgOTN analyser and DUT.

5.1.11.2 Test configuration

The test configuration is shown in Figure 3 above.

5.1.11.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 3. The fibre between fgOTN analyser and DUT is longer than 1 km and identical in both directions.

2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT. Configure TCM monitoring at level TCMi to OPERATIONAL. Select the same TCMi level on fgOTN analyser and DUT.

3) On the EMS: request a bi-directional DM on the TCMi sublayer of the configured fgODUflex.

4) On the fgOTN analyser: request a bi-directional DM on the TCMi sublayer of the configured fgODUflex.

NOTE: fgODUflex supports two TCM levels TCMi.

Repeat the test for each TCMi level supported by the DUT.

5.1.11.4 Expected result

In step 3, the EMS shall report a delay value T_1.

In step 4, the fgOTN analyser shall report a delay value T_2. Verify that T_1 equals T_2.

5.2 fgOTN client services and performance test cases

5.2.1 fgOTN Test case #2.1: STM-N clients

5.2.1.1 Test purpose

This test case verifies the STM-N client services.

5.2.1.2 Test configuration

The test configuration is shown in Figure 4.

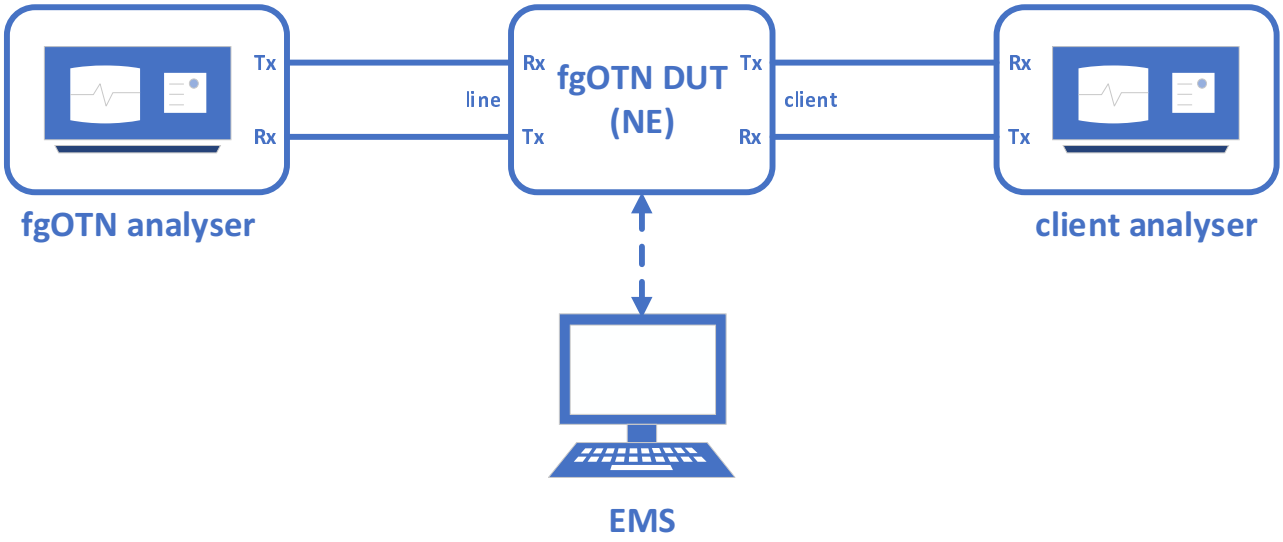


Figure 4: client services test configuration

The test configuration in Figure 4 is reused in multiple test cases below.

5.2.1.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 4. The client analyser is an SDH analyser. The interface between the SDH client analyser and the DUT is an STM-N. The OTN analyser is set to transparent mode (monitor only).

2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT. Configure an STM-N CBR service on the DUT: STM-N → fgODUflex → ODUk.

3) On the fgOTN analyser: check the service mapping.

4) Disconnect the STM-N input on the DUT.

5) Reconnect the STM-N input on the DUT.

Repeat the test for each STM-N level supported by the DUT.

NOTE 1: fgOTN supports STM-1 and STM-4 clients.

Repeat the test for each ODUk server supported by the DUT.

NOTE 2: fgODUflex signals can be multiplexed into multiple ODUk server signals: ODU0, ODU1, ODU2 and ODUflex(p).

5.2.1.4 Expected result

In step 2, the client analyser shall not show any errors. The fgOTN analyser shall not show any errors. The EMS shall not show any errors.

In step 3, the fgOTN analyser shall display the PT value corresponding to the chosen STM-N over fgODUflex service.

In step 4, the fgOTN analyser shall detect the Client Signal Fail (CSF) of the configured fgODUflex set. The EMS shall raise a CSF alarm on the configured fgODUflex.

In step 5, the CSF of the configured fgODUflex shall be cleared on the fgOTN analyser. The EMS shall clear the CSF alarm on the configured fgODUflex.

5.2.2 fgOTN Test case #2.2: STM-N delay and jitter/wander

5.2.2.1 Test purpose

This test case verifies the delay and jitter/wander performance of STM-N clients.

5.2.2.2 Test configuration

The test configuration is shown in Figure 5.

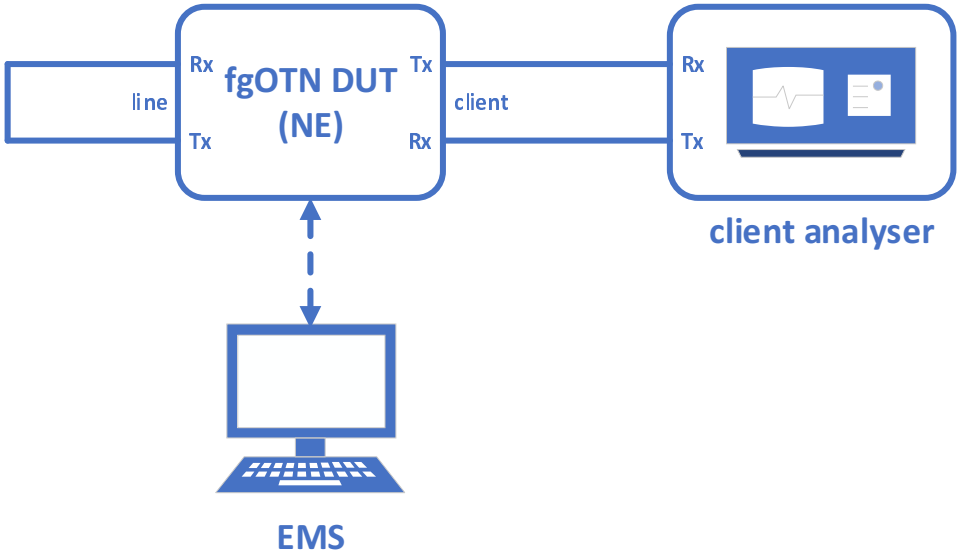


Figure 5: client delay and jitter test configuration

The test configuration in Figure 5 is reused in multiple test cases below.

5.2.2.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 5. The client analyser is an SDH analyser. The interface between the SDH client analyser and the DUT is an STM-N. Perform a fibre loopback on the OTN interface on the line side of DUT.

2) Configure a fgODUflex path between the looped OTN port and the SDH client port connected to the SDH analyser on the DUT. Configure an STM-N CBR service on the DUT: STM-N → fgODUflex → ODUk.

3) Measure the delay of the STM-N client using the SDH analyser.

4) Measure the input jitter and wander tolerance of the STM-N client using the SDH analyser.

5) Measure the generated output jitter of the STM-N client using the SDH analyser.

Repeat the test for each STM-N level supported by the DUT.

NOTE 1: fgOTN supports STM-1 and STM-4 clients.

Repeat the test for each ODUk server supported by the DUT.

NOTE 2: fgODUflex signals can be multiplexed into multiple ODUk server signals: ODU0, ODU1, ODU2 and ODUflex(p).

5.2.2.4 Expected result

In step 3, record the delay on the SDH analyser.

NOTE: The delay requirement depends on the application. Since the application is not specified, the delay is just measured.

In step 4, the input jitter and wander tolerance on the STM-N interface shall comply with the jitter and wander tolerance for STM-N input ports specified in Recommendation ITU-T G.825 [6].

In step 5, the generated output jitter on the STM-N interface shall comply with the ODCp jitter generation as specified in Recommendation ITU-T G.8251 [7].

5.2.3 fgOTN Test case #2.3: Ethernet clients

5.2.3.1 Test purpose

This test case verifies the Ethernet client services.

5.2.3.2 Test configuration

The test configuration is shown in Figure 4 above.

5.2.3.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 4. The client analyser is an Ethernet analyser. The interface between the Ethernet analyser and the DUT is an Ethernet interface. The OTN analyser is set to transparent mode (monitor only).

2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT. Configure an Ethernet service on the DUT: Ethernet → fgODUflex → ODUk.

3) On the fgOTN analyser: check the service mapping.

4) Use the Ethernet analyser to transmit Ethernet traffic. Set the bandwidth ratio to 100 % and the packet lengths to 64 bytes, 128 bytes, 256 bytes, 512 bytes, 1 024 bytes, 1 280 bytes, and 1 518 bytes. Perform the test for 5 minutes and check the traffic transmitted and received.

Repeat the test for each type of Ethernet PHY supported by the DUT.

Repeat the test for each ODUk server supported by the DUT.

NOTE: fgODUflex signals can be multiplexed into multiple ODUk server signals: ODU0, ODU1, ODU2 and ODUflex(p).

5.2.3.4 Expected result

In step 2, the client analyser shall not show any errors. The fgOTN analyser shall not show any errors. The EMS shall not show any errors.

In step 3, the fgOTN analyser shall display the PT value corresponding to the chosen Ethernet over fgODUflex service.

In step 4, the Ethernet analyser shall show that the rates of received and sent packets are the same, no packet loss occurs, and no alarm is generated.

5.2.4 fgOTN Test case #2.4: Ethernet delay

5.2.4.1 Test purpose

This test case verifies the delay performance of Ethernet clients.

5.2.4.2 Test configuration

The test configuration is shown in Figure 5 above.

5.2.4.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 5. The client analyser is an Ethernet analyser. The interface between the Ethernet analyser and the DUT is an Ethernet PHY. Perform a fibre loopback on the OTN interface on the line side of DUT.

2) Configure a fgODUflex path between the looped OTN port and the Ethernet client port connected to the Ethernet analyser on the DUT. Configure an Ethernet service on the DUT: Ethernet → fgODUflex → ODUk.

3) Enable delay measurement on the Ethernet analyser. Record the delay for the packet lengths of 64 bytes, 128 bytes, 256 bytes, 512 bytes, 1 024 bytes, 1 280 bytes, and 1 518 bytes.

Repeat the test for each type of Ethernet PHY supported by the DUT.

Repeat the test for each ODUk server supported by the DUT.

NOTE: fgODUflex signals can be multiplexed into multiple ODUk server signals: ODU0, ODU1, ODU2 and ODUflex(p).

5.2.4.4 Expected result

In step 3, record the delay on the Ethernet analyser.

NOTE: The delay requirement depends on the application/Ethernet service. Since the Ethernet service is not specified, the delay is just measured.

5.2.5 fgOTN Test case #2.5: VC-n clients

5.2.5.1 Test purpose

This test case verifies the VC-n client services.

5.2.5.2 Test configuration

The test configuration is shown in Figure 4 above.

5.2.5.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 4. The client analyser is an SDH analyser. The interface between the SDH client analyser and the DUT is an STM-N. The OTN analyser is set to transparent mode (monitor only).

2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT. Configure one or more VC-n services on the DUT: VC-n → fgODUflex → ODUk.

3) On the fgOTN analyser: check the service mapping.

4) Disconnect the STM-N input on the DUT.

5) Reconnect the STM-N input on the DUT.

Repeat the test for each VC-n level supported by the DUT.

NOTE 1: fgOTN supports VC-12 and VC-4 clients.

Repeat the test for each ODUk server supported by the DUT.

NOTE 2: fgODUflex signals can be multiplexed into multiple ODUk server signals: ODU0, ODU1, ODU2 and ODUflex(p).

5.2.5.4 Expected result

In step 2, the client analyser shall not show any errors on the configured VC-n. The fgOTN analyser shall not show any errors. The EMS shall not show any errors.

In step 3, the fgOTN analyser shall display the PT value corresponding to the chosen VC-n over fgODUflex service.

In step 4, the fgOTN analyser shall detect the CSF of the configured fgODUflex set. The EMS shall raise a CSF alarm on the configured fgODUflex.

In step 5, the CSF of the configured fgODUflex shall be cleared on the fgOTN analyser. The EMS shall clear the CSF alarm on the configured fgODUflex.

5.2.6 fgOTN Test case #2.6: E1 clients

5.2.6.1 Test purpose

This test case verifies the E1 client services.

5.2.6.2 Test configuration

The test configuration is shown in Figure 4 above.

5.2.6.3 Test procedure

The test procedure is as follows:

1) Connect the DUT according to the test configuration in Figure 4. The client analyser is a PDH analyser. The interface between the PDH client analyser and the DUT is an E1. The OTN analyser is set to transparent mode (monitor only).

2) Configure a fgODUflex path between the OTN port connected to the fgOTN analyser and the DUT. Configure one or more E1 services on the DUT: E1 → fgODUflex → ODUk.

3) On the fgOTN analyser: check the service mapping.

4) Disconnect (one of) the E1 input(s) on the DUT.

5) Reconnect the E1 input on the DUT.

Repeat the test: once for a single E1 client and once for the maximum number of E1 clients per fgODUflex supported by the DUT.

NOTE 1: fgOTN supports up to four E1 clients per fgODUflex.

Repeat the test for each ODUk server supported by the DUT.

NOTE 2: fgODUflex signals can be multiplexed into multiple ODUk server signals: ODU0, ODU1, ODU2 and ODUflex(p). Repeat the test for each ODUk server supported by the DUT.

5.2.6.4 Expected result

In step 2, the client analyser shall not show any errors on the configured E1(s). The fgOTN analyser shall not show any errors. The EMS shall not show any errors.

In step 3, the fgOTN analyser shall display the PT value corresponding to the E1 over fgODUflex service.

In step 4, the fgOTN analyser shall detect the CSF of the configured fgODUflex set. The EMS shall raise a CSF alarm on the configured fgODUflex.

In step 5, the CSF of the configured fgODUflex shall be cleared on the fgOTN analyser. The EMS shall clear the CSF alarm on the configured fgODUflex.

5.3 fgOTN protection test cases

5.3.1 fgOTN Test case #3.1: SNC/I protection

5.3.1.1 Test purpose

This test case verifies the 1+1 SNC/I protection functionality of the fgOTN equipment.

NOTE: In SNC/I the protection switch criteria are provided by the server layer. SNC/I is limited to a single server trail for both the working and protection entity.

5.3.1.2 Test configuration

The test configuration is shown in Figure 6.

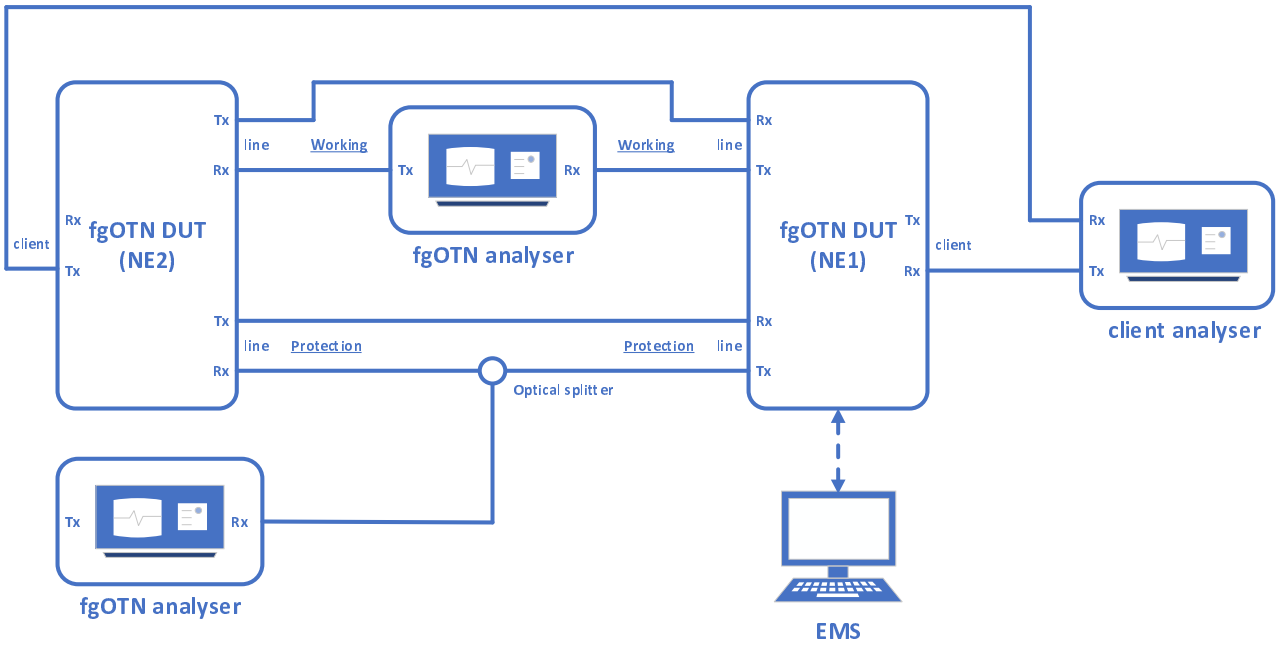


Figure 6: fgODUflex protection test configuration

The test configuration in Figure 6 is reused in multiple test cases below.

5.3.1.3 Test procedure

The test procedure is as follows:

1) Connect the two DUTs (NE1 and NE2) according to the test configuration in Figure 6. The line interfaces between the NEs are OTUks. The fgOTN analyser on the working link is set to transparent mode (monitor only).

2) Configure a SNC/P protected fgODUflex path between NE1 and NE2: client → protected fgODUflex → ODUk → OTUk. Configure the SNC/P protection group for inherent monitoring (SNC/I), 1+1 uni-directional revertive mode. Configure the client over fgODUflex service on both the DUTs and the fgOTN analyser.

3) Create a server alarm on the working link to trigger a protection switch.

4) Clear the server alarm on the working link to trigger the revertive switch.

5) Create an fgODUflex alarm on the working link.

6) Clear the fgODUflex alarm on the working link.

7) Run external commands (Force Switch, Manual Switch, Lockout of Protection) on the EMS and clear the external command.

Repeat steps 3 and 4 for various signal fail and signal degrade triggers. Use the fgOTN analyser on the working link to create LOF, LOM, AIS and TIM alarms on the OTUk, LOFLOM, AIS, LCK, OCI and TIM alarms on the ODUk and LOF and LOM alarms on the fgODUflex. Remove the fibre input on NE2 on the working line to create an LOS alarm.

NOTE 1: The above server alarms result in a Server Signal Fail (SSF) at the fgODUflex triggering an SNC/I protection switch.

Repeat steps 5 and 6 for various fgODUflex alarms. Use the fgOTN analyser on the working link to create, AIS, LCK, OCI and TIM alarms on the fgODUflex.

NOTE 2: The above fgODUflex alarms will not trigger an SNC/I protection switch, only server alarms will.

5.3.1.4 Expected result

In step 2, the client analyser shall not show any errors. Both fgOTN analysers shall not show any errors. The EMS shall not show any errors.

In step 3, record the service interruption time on the client analyser. The EMS shall report the corresponding alarm on NE2 and report a protection switch on the protection group. The EMS shall report the state of the protection group: Signal Degrade (SD) on working on NE2 in case of DEG alarms, Signal Fail (SF) on working on NE2 for all other alarms. The protection switch only occurred in the direction of NE1 to NE2.

In step 4, record the service interruption time on the client analyser. The EMS shall clear the corresponding alarm on NE2 and report a protection switch on the protection group. The EMS shall report the state of the protection group as no request.

In step 5, the EMS shall report the corresponding alarm on NE2 and shall not report any protection switch. The EMS shall report the state of the protection group as no request. Traffic on the client analyser shall be interrupted.

In step 6, the EMS shall clear the corresponding alarm on NE2 and shall not report any protection switch. The EMS shall report the state of the protection group as no request. Traffic on the client analyser shall be restored.

In step 7, record the service interruption times on the client analyser. The EMS shall report the corresponding state of the protection group. The protection switch only occurred in the direction of NE1 to NE2.

The service interruption times in steps 3 through 4 and step 7 shall be less than 50 ms for service rate of 50 Mbit/s or above.

NOTE: Service interruption times for service rates below 50 Mbit/s are for further study.

5.3.2 fgOTN Test case #3.2: SNC/N protection

5.3.2.1 Test purpose

This test case verifies the 1+1 SNC/N protection functionality of the fgOTN equipment.

NOTE: In SNC/N the protection switch criteria are provided by non-intrusively monitoring either the end-to-end (PM) or sublayer (TCMi) overhead.

5.3.2.2 Test configuration

The test configuration is shown in Figure 6 above.

5.3.2.3 Test procedure

The test procedure is as follows:

1) Connect the two DUTs (NE1 and NE2) according to the test configuration in Figure 6. The line interfaces between the NEs are OTUks. The fgOTN analyser on the working link is set to transparent mode (monitor only).

2) Configure a SNC/P protected fgODUflex path between NE1 and NE2: client → protected fgODUflex → ODUk → OTUk. Configure the SNC/P protection group for non-intrusive monitoring (SNC/N), 1+1 uni-directional revertive mode. Configure the client over fgODUflex service on both the DUTs and the fgOTN analyser. Configure an TTI string on the fgODUflex. Select the same monitoring level on fgOTN analyser and DUT.

3) Create an alarm on the working link to trigger a protection switch.

4) Clear the alarm on the working link to trigger the revertive switch.

5) Run external commands (Force Switch, Manual Switch, Lockout of Protection) on the EMS and clear the external command.

Repeat steps 3 and 4 for various signal fail and signal degrade triggers. Use the fgOTN analyser on the working link to create, LOF, LOM, AIS, LCK, OCI and TIM alarms on the fgODUflex. Remove the fibre input on NE2 on the working line to create an LOS alarm.

NOTE 1: The LOF and LOM alarms on the fgODUflex as well as the LOS result in a Server Signal Fail (SSF) at the fgODUflex. The AIS, LCK, OCI and TIM alarms and the SSF on the fgODUflex trigger an SNC/N protection switch.

NOTE 2: fgODUflex supports three monitoring levels: PM and two TCM levels TCMi.

Repeat the test for the PM level and each TCMi level supported by the DUT.

5.3.2.4 Expected result

In step 2, the client analyser shall not show any errors. Both fgOTN analysers shall not show any errors. The EMS shall not show any errors. On NE2 the non-intrusive monitors on both the working and the protection path shall report the same TTI string.

In step 3, record the service interruption time on the client analyser. The EMS shall report the corresponding alarm on NE2 and report a protection switch on the protection group. The EMS shall report the state of the protection group: Signal Degrade (SD) on working on NE2 in case of DEG alarms, Signal Fail (SF) on working on NE2 for all other alarms. The protection switch only occurred in the direction of NE1 to NE2.

In step 4, record the service interruption time on the client analyser. The EMS shall clear the corresponding alarm on NE2 and report a protection switch on the protection group. The EMS shall report the state of the protection group as no request.

In step 5, record the service interruption times on the client analyser. The EMS shall report the corresponding state of the protection group. The protection switch only occurred in the direction of NE1 to NE2.

The service interruption times in steps 3 through 5 shall be less than 50 ms for service rate of 50 Mbit/s or above.

NOTE: Service interruption times for service rates below 50 Mbit/s are for further study.

5.3.3 fgOTN Test case #3.3: SNC/S protection

5.3.3.1 Test purpose

This test case verifies the 1+1 SNC/S protection functionality of the fgOTN equipment.

NOTE: In SNC/S the protection switch criteria are provided by terminating and monitoring the sublayer (TCMi) overhead.

5.3.3.2 Test configuration

The test configuration is shown in Figure 6 above.

5.3.3.3 Test procedure

The test procedure is as follows:

1) Connect the two DUTs (NE1 and NE2) according to the test configuration in Figure 6. The line interfaces between the NEs are OTUks. The fgOTN analyser on the working link is set to transparent mode (monitor only).

2) Configure a SNC/P protected fgODUflex path between NE1 and NE2: client → protected fgODUflex → ODUk → OTUk. Configure the SNC/P protection group for sublayer monitoring (SNC/S), 1+1 uni-directional revertive mode. Configure the client over fgODUflex service on both the DUTs and the fgOTN analyser. Configure a different TTI string at the TCMi sublayer on the working and the protection path. Select the same TCMi level on fgOTN analyser and DUT.

3) Create an alarm on the working link to trigger a protection switch.

4) Clear the alarm on the working link to trigger the revertive switch.

5) Run external commands (Force Switch, Manual Switch, Lockout of Protection) on the EMS and clear the external command.

Repeat steps 3 and 4 for various signal fail and signal degrade triggers. Use the fgOTN analyser on the working link to create LOF, LOM, AIS, LCK, OCI and TIM alarms on the fgODUflex. Remove the fibre input on NE2 on the working line to create an LOS alarm.

NOTE 1: The LOF and LOM alarms on the fgODUflex as well as the LOS result in a Server Signal Fail (SSF) at the fgODUflex. The AIS, LCK, OCI and TIM alarms and the SSF on the fgODUflex trigger an SNC/S protection switch.

NOTE 2: fgODUflex supports two TCM levels TCMi.

Repeat the test for each TCMi level supported by the DUT.

5.3.3.4 Expected result

In step 2, the client analyser shall not show any errors. Both fgOTN analysers shall not show any errors. The EMS shall not show any errors. On NE2 the TCMi monitors on the working and the protection path shall report the TTI string configured in step 2.

In step 3, record the service interruption time on the client analyser. The EMS shall report the corresponding alarm on NE2 and report a protection switch on the protection group. The EMS shall report the state of the protection group: Signal Degrade (SD) on working on NE2 in case of DEG alarms, Signal Fail (SF) on working on NE2 for all other alarms. The protection switch only occurred in the direction of NE1 to NE2.

In step 4, record the service interruption time on the client analyser. The EMS shall clear the corresponding alarm on NE2 and report a protection switch on the protection group. The EMS shall report the state of the protection group as no request.

In step 5, record the service interruption times on the client analyser. The EMS shall report the corresponding state of the protection group. The protection switch only occurred in the direction of NE1 to NE2.

The service interruption times in steps 3 through 5 shall be less than 50 ms for service rate of 50 Mbit/s or above.

NOTE: Service interruption times for service rates below 50 Mbit/s are for further study.

5.4 fgOTN hitless bandwidth adjustment test cases

5.4.1 fgOTN Test case #4.1: hitless bandwidth adjustment without protection

5.4.1.1 Test purpose

This test case verifies that (unprotected) Ethernet MAC client services support hitless bandwidth increase and decrease, as well as rollback after error conditions.

5.4.1.2 Test configuration

The test configuration is shown in Figure 7.

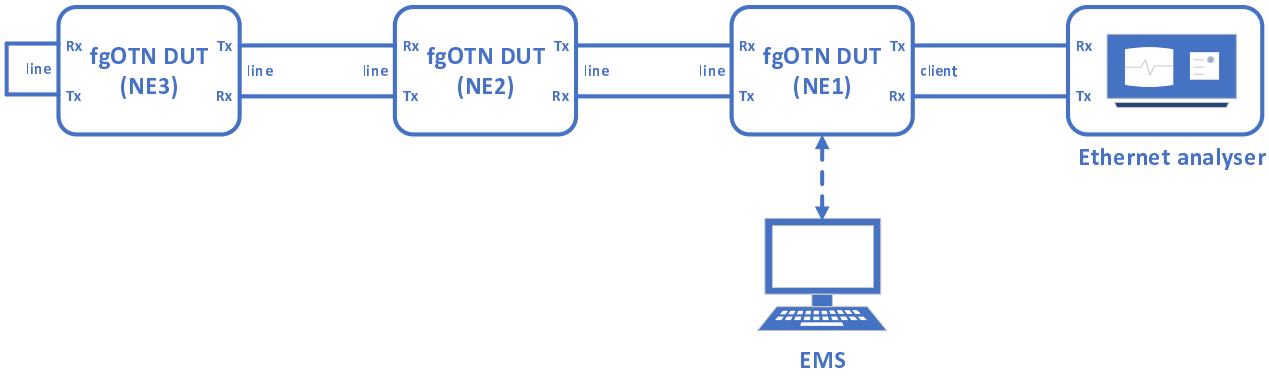


Figure 7: fgOTN hitless bandwidth adjustment without protection test configuration

5.4.1.3 Test procedure

The test procedure is as follows:

1) Connect the three DUTs (NE1, NE2 and NE3) according to the test configuration in Figure 7. The interface between the Ethernet analyser and NE1 is a 1000BASE-R Ethernet interface. The line interfaces between the NEs are OTUks. Perform a fibre loopback on the OTN interface on the line side of NE3.

2) Configure a fgODUflex path between the OTN line port on NE1, pass-through NE2 (one port connected to NE1, the other to NE3) and pass-through NE3 (one port connected to NE2, the other to the loopback). Configure a 10 Mbit/s Ethernet MAC service on that fgODUflex: Ethernet → fgODUflex → ODUk. Configure a 10 Mbit/s Ethernet traffic flow on the Ethernet analyser.

3) Change the fgODUflex bandwidth on NE1 from 10 Mbit/s to 100 Mbit/s.

4) Change the traffic rate on the Ethernet analyser from 10 Mbit/s to 100 Mbit/s.

5) Change the fgODUflex bandwidth on NE1 from 100 Mbit/s to 1 000 Mbit/s.

6) Change the traffic rate on the Ethernet analyser from 100 Mbit/s to 1 000 Mbit/s.

7) Change the traffic rate on the Ethernet analyser from 1 000 Mbit/s to 10 Mbit/s.

8) Change the fgODUflex bandwidth on NE1 from 1 000 Mbit/s to 10 Mbit/s.

9) Change the traffic rate on the Ethernet analyser from 10 Mbit/s to 50 Mbit/s.

10) Change the traffic rate on the Ethernet analyser from 50 Mbit/s to 10 Mbit/s.

11) Disconnect the fibre from NE2 to NE3. Change the fgODUflex bandwidth on NE1 from 10 Mbit/s to 100 Mbit/s.

12) Reconnect the fibre from NE2 to NE3.

13) Configure additional fgODUflex services on the link between NE2 and NE3 until the ODUk is filled with less than 90 Mbit/s available capacity. Change the fgODUflex bandwidth on NE1 from 10 Mbit/s to 100 Mbit/s.

5.4.1.4 Expected result

In step 2, the Ethernet analyser shall show that the rates of received and sent packets are the same, no packet loss occurs, and no alarm is generated. The EMS shall not show any errors.

In steps 3 through 8, the Ethernet analyser shall show that the rates of received and sent packets are the same, no packet loss occurs, and no alarm is generated. The EMS shall not show any errors.

In step 9, the Ethernet analyser shall show that the rates of received packets is only 10 Mbit/s and a packet loss of 80 % occurs. The EMS shall not show any errors.

In step 10, the Ethernet analyser shall show that the rates of received and sent packets are the same, no packet loss occurs, and no alarm is generated. The EMS shall not show any errors.

In step 11, the Ethernet analyser shall show that there is no packet received. The EMS shall report a LOS alarm on NE3 and shall report the fgODUflex bandwidth increase failed and was rolled back to 10 Mbit/s.

In step 12, the Ethernet analyser shall show that the rates of received and sent packets are the same, no packet loss occurs, and no alarm is generated. The EMS shall not show any errors.

In step 13, the Ethernet analyser shall show that the rates of received and sent packets are the same, no packet loss occurs, and no alarm is generated. The EMS shall report the fgODUflex bandwidth increase failed and was rolled back to 10 Mbit/s due to a lack of resources.

5.4.2 fgOTN Test case #4.2: hitless bandwidth adjustment in protection scenarios

5.4.2.1 Test purpose

This test case verifies that protected Ethernet MAC client services support hitless bandwidth increase and decrease, as well as rollback after error conditions.

5.4.2.2 Test configuration

The test configuration is shown in Figure 8.

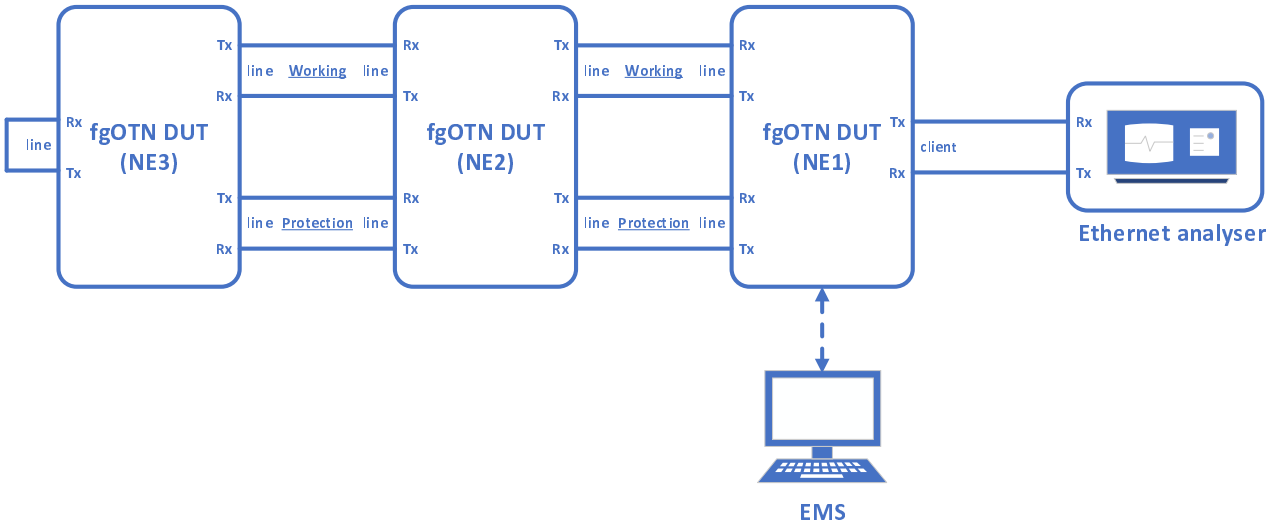


Figure 8: test configuration for hitless bandwidth adjustment in protection scenarios

5.4.2.3 Test procedure

The test procedure is as follows:

1) Connect the three DUTs (NE1, NE2 and NE3) according to the test configuration in Figure 8. The interface between the Ethernet analyser and NE1 is a 1000BASE-R Ethernet interface. The line interfaces between the NEs are OTUks. Perform a fibre loopback on the OTN interface on the line side of NE3.

2) Configure a SNC/S protected fgODUflex path between the OTN line port on NE1, pass-through NE2 (one port connected to NE1, the other to NE3) and pass-through NE3 (one port connected to NE2, the other to the loopback). Configure the SNC/S protection group bi-directional revertive. Configure a 100 Mbit/s Ethernet MAC service on that fgODUflex: Ethernet → fgODUflex → ODUk. Configure a 100 Mbit/s Ethernet traffic flow on the Ethernet analyser.

3) Disconnect the fibre with the working path of the protection group from NE2 to NE3.

4) Reconnect the fibre with the working path of the protection group fibre from NE2 to NE3 and wait until the revertive switch occurs.

5) Change the fgODUflex bandwidth on NE1 from 100 Mbit/s to 500 Mbit/s.

6) Change the traffic rate on the Ethernet analyser from 100 Mbit/s to 500 Mbit/s.

7) Disconnect the fibre with the working path of the protection group from NE2 to NE3.

8) Reconnect the fibre with the working path of the protection group from NE2 to NE3 and wait until the revertive switch occurs.

9) Disconnect the fibre with the protection path of the protection group from NE2 to NE3.

10) Change the fgODUflex bandwidth on NE1 from 500 Mbit/s to 1 000 Mbit/s.

11) Reconnect the fibre with the protection path of the protection group from NE2 to NE3.

12) Change the fgODUflex bandwidth on NE1 from 500 Mbit/s to 1 000 Mbit/s.

13) Change the traffic rate on the Ethernet analyser from 500 Mbit/s to 1 000 Mbit/s.

5.4.2.4 Expected result

In step 2, the Ethernet analyser shall show that the rates of received and sent packets are the same, no packet loss occurs, and no alarm is generated. The EMS shall not show any errors.

In step 3, record the service interruption time on the Ethernet analyser. The EMS shall report a LOS alarm on NE3 and report a protection switch on the SNC/S protection group. The EMS shall report the state of the protection group as Signal Fail (SF) on working.

In step 4, record the service interruption time on the Ethernet analyser. The EMS shall clear the LOS alarm on NE3 and report a protection switch on the SNC/S protection group. The EMS shall report the state of the protection group as No Request (NR).

In steps 5 through 6, the Ethernet analyser shall show that the rates of received and sent packets are the same, no packet loss occurs, and no alarm is generated. The EMS shall not show any errors.

In step 7, record the service interruption time on the Ethernet analyser. The EMS shall report a LOS alarm on NE3 and report a protection switch on the SNC/S protection group. The EMS shall report the state of the protection group as Signal Fail (SF) on working.

In step 8, record the service interruption time on the Ethernet analyser. The EMS shall clear the LOS alarm on NE3 and report a protection switch on the SNC/S protection group. The EMS shall report the state of the protection group as No Request (NR).

In step 9, the Ethernet analyser shall show that the rates of received and sent packets are the same, no packet loss occurs, and no alarm is generated. The EMS shall not show any errors.

In step 10, the Ethernet analyser shall show that the rates of received and sent packets are the same, no packet loss occurs, and no alarm is generated. The EMS shall report that the fgODUflex bandwidth increase failed and was rolled back to 500 Mbit/s due to the faulty protection path.

In steps 11 through 13, the Ethernet analyser shall show that the rates of received and sent packets are the same, no packet loss occurs, and no alarm is generated. The EMS shall not show any errors.

5.5 fgOTN cross-connect test cases

5.5.1 fgOTN Test case #5.1: cross-connect capacity

5.5.1.1 Test purpose

This test case verifies the cross-connection capacity of the fgODUflex in the fgOTN equipment and verifies the number of fgODUflex paths supported by the cross-connect.

5.5.1.2 Test configuration

The test configuration is shown in Figure 9.

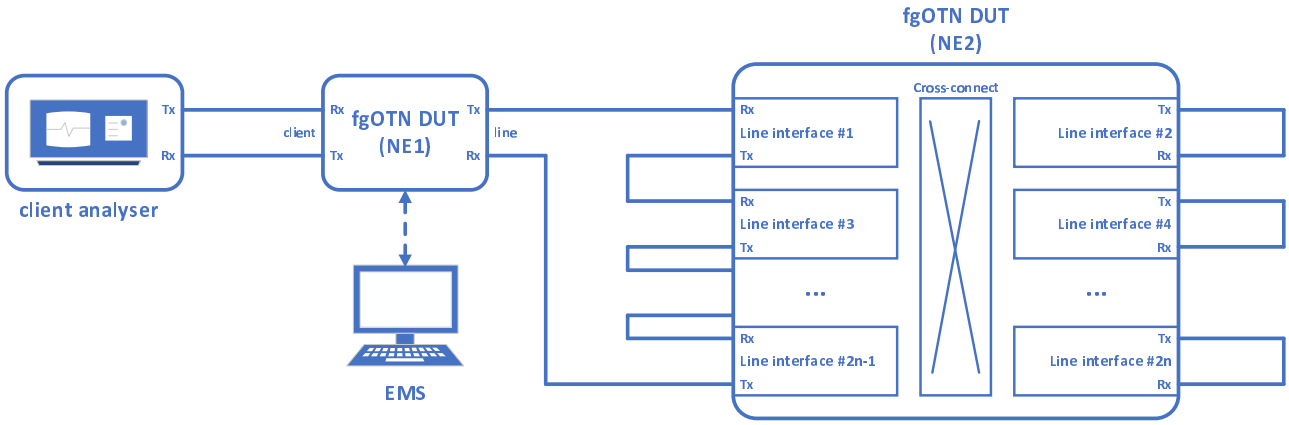


Figure 9: cross-connect capacity test configuration

5.5.1.3 Test procedure

The test procedure is as follows:

1) Connect the two DUTs (NE1 and NE2) according to the test configuration in Figure 9. The line interfaces between the NEs are OTUks. Connect line interfaces on NE2 in serpentine mode as shown in Figure 9. The client analyser is an Ethernet analyser.

2) On NE1, configure multiple client interfaces with Ethernet services: Ethernet → fgODUflex.

3) On both NE1 and NE2, configure the maximum number m of fgODUflex paths based on the capabilities of the line interface: fgODUflex [→ ODUj] → ODUk → OTUk. Configure the bandwidth for each fgODUflex to reach the maximum capacity of the line interface.

NOTE: The square brackets [] indicate the optional part of the multiplexing hierarchy of the interface.

4) On NE1, configure bi-directional connections from the client interfaces to the line interface in the cross-connect matrix. On NE2, configure bi-directional connections between line interface #2n-1 to line interface #2n. In case the number of line interfaces on NE2 is odd, configure fgODUflex loopbacks on the last line interface.

5) Use the Ethernet analyser to transmit Ethernet traffic. Set the bandwidth ratio to 100 %. Perform the test for 5 minutes and check the traffic transmitted and received.

6) Remove a fibre on NE2.

7) Reconnect the fibre on NE2.

Repeat steps 6 to 7 for each of the fibres on NE2.

5.5.1.4 Expected result

In step 5, the client analyser shall not show any errors and no data loss.

In step 6, the client analyser shall show 100 % traffic loss.

In step 7, the client analyser shall not show any errors and no data loss.

5.5.2 fgOTN Test case #5.2: cross-connect delay

5.5.2.1 Test purpose

This test case verifies the transfer delay of an fgODUflex through the cross-connect of the fgOTN equipment.

5.5.2.2 Test configuration

The test configuration is shown in Figure 10.

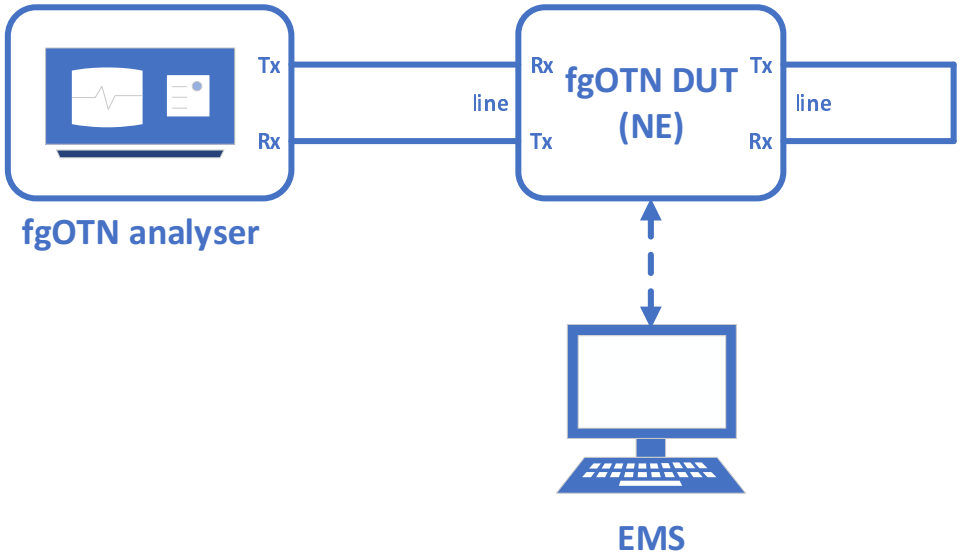


Figure 10: Cross-connect delay test configuration

5.5.2.3 Test procedure

The test procedure is as follows:

1) Perform a system loopback (internal transmit to receive) on the fgOTN analyser and measure the inherent bi-directional delay of the analyser.

2) Connect the DUT according to the test configuration in Figure 10. Perform a fibre loopback on the OTN interface on the line side OTUk port of the DUT that is not connected to the analyser.

3) Configure a bi-directional fgODUflex cross-connect between the OTN port connected to the fgOTN analyser and the DUT and the OTN port connected to the fibre loopback.

4) On the fgOTN analyser: request a bi-directional DM on the PM layer of the configured fgODUflex.

Repeat steps 3 and 4 for different fgODUflex granularities such as 10 Mbit/s, 100 Mbit/s, and 1 000 Mbit/s and the different fgODUflex mapping hierarchies supported by the DUT.

NOTE: In TDM technologies traffic load should not affect the cross-connect transfer delay. The effects of traffic load are for further study.

5.5.2.4 Expected result

In step 1, record the inherent delay value T0 of the analyser.

In step 4, record the delay value T_i. Calculate the device transfer delay $T_t = (T_i – T_0)/2$.

5.5.3 fgOTN Test case #5.3: fgODUflex container capacity

5.5.3.1 Test purpose

This test case verifies the fgODUflex container capacity of line ports of the fgOTN equipment.

5.5.3.2 Test configuration

The test configuration is shown in Figure 11.

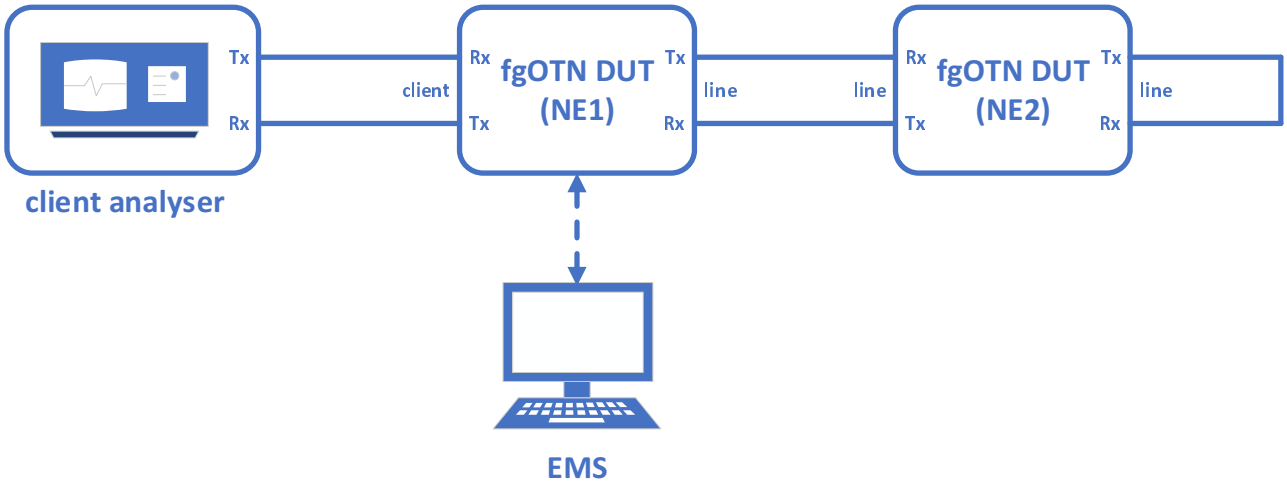


Figure 11: fgODUflex container capacity test configuration

5.5.3.3 Test procedure

The test procedure is as follows:

1) Connect the two DUTs (NE1 and NE2) according to the test configuration in Figure 11. The line interfaces between the NEs are OTUks. Perform a loopback on NE2, either by a loopback of the OTN interface or by fgODUflex loopbacks in the cross-connect of NE2. The client analyser is an Ethernet analyser.

2) On NE1, configure multiple client interfaces with Ethernet services: Ethernet → fgODUflex.

3) On both NE1 and NE2, configure the maximum number of fgODUflex paths based on the capabilities of the line card: fgODUflex [→ ODUj] → ODUk → OTUk.

NOTE 1: The square brackets [] indicate the optional part of the multiplexing hierarchy of the interface.

4) On NE1, configure bi-directional connections from the client interfaces to the line interface in the cross-connect matrix. On NE2, in the fibre loopback case, configure bi-directional connections between both line interfaces in the cross-connect matrix.

5) Use the client analyser to check the service on each of the client interfaces.

6) Remove the loopback fibre on NE2 or delete the fgODUflex loopback in the cross-connect of NE2.

7) Reconnect the loopback fibre on NE2 or re-configure the fgODUflex loopback in the cross-connect of NE2.

Repeat steps 5 to 7 for each of the configured client interfaces.

8) Record the maximum number of fgODUflex supported by the corresponding line card and its OTUk interface.

Repeat the test for each ODUk server and multiplexing option supported by the DUT.

NOTE 2: fgODUflex signals can be multiplexed into multiple ODUk server signals: ODU0, ODU1, ODU2 and ODUflex(p).

NOTE 3: ODUk server signals can be mapped directly into OTUk interface signals or multiplexed into a higher-order ODUk signal: ODU1 and ODU2.

5.5.3.4 Expected result

In step 5, the client analyser shall not show any errors and no data loss.

In step 6, the client analyser shall show 100 % traffic loss.

In step 7, the client analyser shall not show any errors and no data loss.

In step 8, record the number of fgODUflex supported on the line interface.

Annex A (informative): Change history

Date	Version	Information about changes
2025-09-09	V0.0.1	Initial skeleton, F5GTEST(25)004009
2025-11-12	V0.0.2	Editorial review of WI35 early draft, F5GTEST(25)005011 fgODUflex alignment test cases for WI35, F5GTEST(25)005012r1
2026-02-03	V0.0.3	fgODUflex overhead test cases for WI35, F5GTEST(25)006002r2
2026-04-17	V0.0.4	fgOTN client services test cases for WI35, F5GTEST(26)007008r1 Hitless adjustment test cases for WI35, F5GTEST(26)000010r2 Addressed multiple Editor's notes
2026-05-19	V0.0.5	fgODUflex protection test cases for WI35, F5GTEST(26)000014r2 Addressed multiple Editor's notes Editorial updates (UK English, figure numbers)
2026-06-10	V0.0.6	fgODUflex cross connect test cases for WI35, F5GTEST(26)008004r1 fgODUflex OCI and AIS test steps for WI35, F5GTEST(26)008005r1 Test cases overview table for WI35, F5GTEST(26)008006r1 Addressed multiple Editor's notes Clarified SNC/P protection operation Editorial updates (TCMi sublayer, numbered lists,references)
2026-07-29	V0.0.7	Review of WI-35 draft 0.0.6, F5GTEST(26)000022 Responses to F5GTEST(26)000022, F5GTEST(26)000025r1

History

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